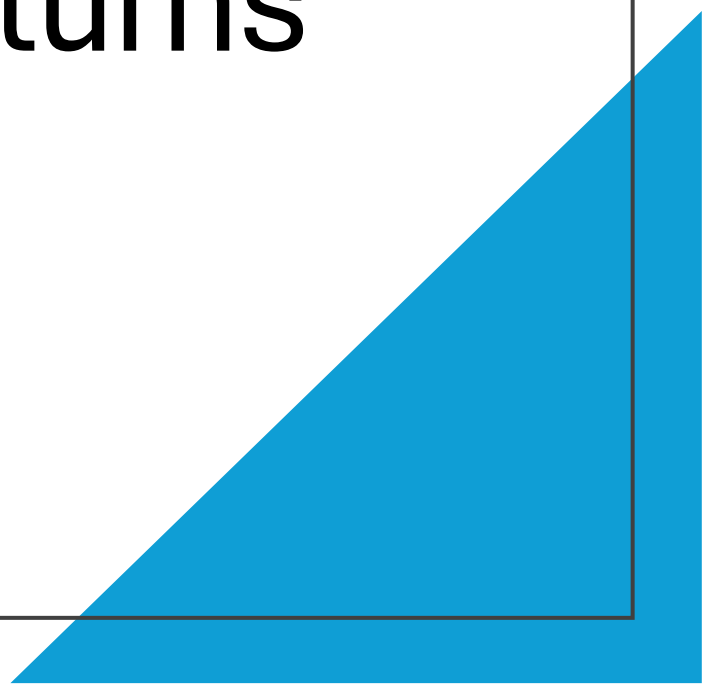


Does Disagreement Among Equity Factors Signal Higher or Lower Market Excess Returns Tomorrow?

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Does Disagreement Among Equity factors Signal Higher or Lower Market Excess Returns?– Context

- **Dataset:** The dataset is the Fama-French 5-Factor daily returns data. It contains information regarding to SMB, HML, RMW, CMA, and Mkt-RF. This information is sourced from the Kenneth French Data Library and the data relates to daily United States Equity Factor returns (stock market risk and return).
- **Machine Learning Models Used:** A Ridge Regression Model was used with regularization. The model is predicting the next day's market return. It attempts to determine whether high or low factor disagreement could be used to determine market performance the following day. Factors used include: SMB, HML, RMW, CMA, and Mkt-RF.
- **Future Implications:**
 - What impact do political and social events have in predicting market performance?
 - How do unexpected, high impact market shocks impact the stability of the factor disagreement signal?
 - How does behavioral bias impact factor disagreement as a determining factor for market performance?

Input and Target Variables

MKT-RF_Lag1	Yesterday's market return minus the risk-free rate.
Factor_Disagreement_5d	The disagreement of the 4 main market factors the last 5 days. Higher number = factors strongly disagree.
market_vol_21d	The volatility of the market the last 21 days.
Factor_minus_market_mom	The markets momentum subtracted from the factor's momentum.
Disagreement_x_vol	Investor disagreement multiplied by market volatility.
SMB (Small Minus Big)	The return of small companies minus big companies.
HML (High Minus Low)	High minus Low. The return of value stocks minus growth stocks.
RMW (Robust Minus Weak)	The returns of unprofitable firms subtracted from profitable firms.
CMA (Conservative Minus Aggressive)	The returns of firms that invest aggressively subtracted from firms that invest conservative.
RF (<i>Risk Free Rate</i>)	The return received with little to no risk.
<u>Target_next_mkt</u>	The market return in the next day. This is the target variable.

Evaluation Metrics Used

RMSE (Root Mean Square Error)	Number that describes how far away the predictions are on average from the actual results. Lower RMSE represents better predictions.
Sharpe	The amount of extra return received for each additional unit of risk. A higher value indicates a better risk-adjusted performance.
Information Coefficient	The effect that the change of one variable has on another changed variable. A positive interaction implies that as the variable increases, the impact of it on the other variable also increases.

Does Disagreement Among Equity factors Signal Higher or Lower Market Excess Returns?– Conclusion

Summary: The purpose of this project was to see if disagreement among equity factors could be used to predict next day market performance. The model produced a negative Sharpe Ratio and Information Coefficient, indicating that the factor disagreement signal had essentially no effect in predicting next-day market returns. This proves that the Ridge Regression Model struggled to extract any meaningful insights from the input data.

Key Takeaways: As stated above, factor disagreement did not predict next-day market returns in this model. The variable 'factor_disagreement_5d' had the largest positive coefficient, indicating that there might be something there to further investigate in the future and carry some significance. Moreover, the 'disagreement_x_vol' had close to no impact as well in the model, revealing that volatility combined with disagreement among equity factors did not add any stress-signal.

Next Steps: In the future, I am looking forward to investigating this matter further. I believe testing more models, such as XGBoost and LSTM could be intriguing to look at. Involving more data, such as inflation reports, GDP growth, and VIX fear index would capture more information in regards to investor and economic activity. Retesting this project with these models and new features could get a much better model performance.